The Integration of Motivation and Volition in Personality Systems Interactions (PSI) Theory

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Abstract

This chapter outlines the relationships between motivation and volition and their integration within personality systems interactions (PSI) theory. The original focus on motivation in contemporary times in the USA (Atkinson, McClelland) and in Germany (Heckhausen) shifted towards volitional processes to explain human behavior and experience. We review how this shift was facilitated by the work of Julius Kuhl in the 1980s, which led to his theory of action control, including the personality disposition of action versus state orientation. PSI theory extended the scope of the work to all personality functioning. This theory postulates seven levels of personality with increasing phylogenetic and ontogenetic complexity (*habits, temperament, affect, coping with stress, motives, cognitive systems, and self-management*), each of which has been the focus of influential theories of personality. We first address these levels and then describe the four mental systems in PSI theory (extension memory, intention memory, object recognition, and intuitive behavior control) together with assumptions about how interactions between these systems are modulated by positive and negative affect to explain experience and behavior. We illustrate afterwards PSI-theory's integrative potential by showing how the Rubicon model and emotion regulation could be conceptualized within this theory. We finally stress the relevance of PSI theory (and of the diagnostic system derived from it) for other theories and for its application in clinical and other contexts.

The Integration of Motivation and Volition in Personality Systems Interactions (PSI) Theory

Motivation refers to the extent to which our behavior is selected, directed, energized, and maintained to satisfy a particular motive, such as affiliation, achievement, or power. The motivational strength in a given moment depends not only on a latent motive disposition but also on situational incentives and the expectation of a successful outcome. *Volition* (self-government) refers to a central executive function in charge of coordinating many different "sub-functions" such as the dominant action tendency, affect, motives, or cognitive processes to reach a particular goal and to shield our behavior from competing external or internal action tendencies. *Personality* refers to the characteristic manner of the interplay between cognitive, emotional, and motivational processes.

In this chapter, we discuss how psychology of motivation, volition, and personality can be integrated by delineating the development of Julius Kuhl's theoretical ideas and empirical research, which culminated in Personality Systems Interactions (PSI) theory (Kuhl, 2000, 2001, 2010). Over the last three decades, there has been a continuous theoretical development in the papers of Julius Kuhl starting with motivation, going on to volition, and ending with personality. This development may not be obvious to colleagues working in the areas of motivation, volition, personality, neuropsychology, or cognitive science because researchers typically tend to see only the contributions to the area they are experts in. We attempt to delineate these developments and to highlight how they are linked to these areas. As such, this chapter not only reviews the historical development of Julius Kuhl's work but also demonstrates how motivation, volition, and personality may be integrated in a single theoretical framework.

In the following, we start by recounting the theoretical roots of PSI theory, which lie in the domains of motivation and volition research. We then discuss the original action control theory (Kuhl, 1984) and the notion of action versus state orientation, as the theoretical

predecessors of PSI theory. After this, we outline PSI theory and consider how it may be applied to an influential conception of volition and to action-state orientation. Finally, we summarize our main conclusions and consider the broad theoretical and applied implications of PSI theory.

From Motivation to Volition

Classical theories of motivation (Atkinson & Feather, 1966; McClelland, Atkinson, Clark, & Lowell, 1953) assume that a human need, like the need to attain success (achievement motive), the need to be with people (affiliation motive), or the need to influence other people (power motive), selects, directs, and energizes behavior. On the one hand, positive affect arises if an opportunity occurs to satisfy the need or if the individual makes progress towards satisfying the need. On the other hand, a need may arouse avoidance tendencies and be associated with negative affect if a situation occurs with the risk of experiencing an aversive state (e.g., task failure, personal rejection, or helplessness). This approach-avoidance dichotomy of human motivation, which has strongly influenced the measurement of motives (McClelland, 1987; Morgan & Murray, 1935; Schultheiss & Brunstein, 2010; Winter, 1991), has been later on complemented with constructs such as mastery versus performance orientation (Dweck, 1986), task- versus ego-involvement (Nicholls, 1984), certainty versus uncertainty orientation (Sorrentino, Short, & Raynor, 1984), or intrinsic versus extrinsic motivation (Deci & Ryan, 1985). Subsequently, motivation accounts have been related to the self or to self-regulatory functions, but the focus has been on cognitive or intentional accounts. That is, they explain motivation according to the individual's self-attributions, thoughts, desires, or intentions. Famous examples are attribution theory (Weiner, 1980), symbolic self-completion theory (Wicklund & Gollwitzer, 1982), self-efficacy theory (Bandura, 1986), self-discrepancy theory (Higgins, 1987) and self-determination theory (Ryan & Deci, 2000).

Theories of motivation with a focus on cognitive constructs have also been subsumed under the label “Expectancy x Value” models (Atkinson, 1957; Feather, 1961; Vromm, 1964; Weiner, 1980). In general, these models assume that each volitional action results from a rational decision process. When an individual is confronted with a new situation, the actor considers the available goals and chooses the goal with the highest value expected. Subsequently, the individual considers the action options that are available to pursue the goal, and again chooses the action alternative with the highest expected value to reach that goal. These models have been criticized because of their exclusive reliance on cognitive constructs and conscious decision making, which give an idealized and unrealistic account of how human action and decision making take place (Heckhausen, 1977; Heckhausen & Kuhl, 1985; Kuhl, 1983; Tversky & Kahneman, 1975).

In the 1980s, Kuhl concluded that volitional factors are needed to explaining how motivation becomes translated into action. To do this, Kuhl went back to proposals of early German motivational and will psychology theorists, like Kurt Lewin and Narziß Ach. In the book based on his “Habilitation”: *Motivation, Conflict, and Action Control*, written in 1983, he discusses in detail these issues: Motivational theory of Kurt Lewin, expectancy-value models of motivation, dynamic theory of motivation from Atkinson and Birch, and theory of action control, including Narziß Ach’s concept on volition. The above chapters gave an overview of the state of the art in motivation and volition at the time, including a historical review of the controversy between Kurt Lewin and Narziß Ach. Whereas Lewin favored a motivational perspective, Ach favored a volitional approach to explain complex human action. Kuhl distinguishes in the book between “selection motivation” and “enactment motivation” (realization motivation), which was the foundation of further developments of volitional psychology, including the Rubicon model. The book ends with a theory of action control, including the individual difference of action versus state orientation.

One important step in the theoretical transition from motivation to volition was given in the chapter of Heckhausen and Kuhl (1985), “From wishes to action…” in which key ideas underlying the Rubicon model were introduced. The Rubicon model of action phases was further developed by Heckhausen (1987) and by Heckhausen and Gollwitzer (1987), distinguishing motivational and volitional processes to explain human action. The four action phases are (1) pre-decisional, (2) pre-actional, (3) actional, and (4) post-actional. This theory attempts to explain, respectively: (a) How do people select their goals? (b) How do they plan the execution of those goals? (c) How do they enact those goals? and (d) How do they evaluate their efforts to accomplish a specific goal? (Achtzinger & Gollwitzer, 2008). The pre-decisional phase is motivational, with a *deliberative mindset*, in which expectancy-value considerations take place and a host of alternative actions are cognitively evaluated. Once the “Rubicon” is crossed with the formation of a decision and corresponding intention, two volitional phases come into play: Pre-actional and actional, with an *implemental mindset*, in which a narrow set of alternatives related to the intention is considered. In the pre-actional phase, planning dominates whereas in the actional phase implementation of the intention takes place. After the intention is achieved (or disengaged), it is deactivated, and in the post-actional phase a deliberative evaluative mindset takes place, which again is considered motivational (Heckhausen & Gollwitzer, 1987).

Kuhl’s Theory of Action Control

While Heckhausen and Gollwitzer were elaborating the Rubicon model, Kuhl was working to develop a more comprehensive theory of action control (Kuhl, 1984, 1987). The theory of action control was initially developed by Kuhl distinguishing two types of action control, *passive* and *active* (Kuhl, 1987, pp. 286-289). Passive action control supports the motivationally strongest action tendency. This mode of control is mostly dependent on automatic attentional mechanisms and it is carried out through well-learned behavioral routines. Active action control, on the other hand, can support the enactment of subordinate,

motivationally weak intentions through self-regulatory strategies: (a) selective attention, (b) encoding control, (c) emotion control, (d) motivation control, (e) environment control, and (f) parsimony of information processing (Kuhl, 1984).

Within Kuhl's theory of action control, there is much attention to individual differences in volitional control, which are captured by the notion of *action versus state orientation* (Kuhl & Beckmann, 1994). There are three different forms, which are measured with the action control scale (Kuhl, 1994b): First, action versus state orientation (a) after threat or failure, (b) demand-related (decision and initiative), and (c) after successful performance. Failure- or threat-related state orientation describes individuals who have persistent and uncontrollable negative emotional states after being exposed to aversive events. They focus their attention on a past, present, or future state instead of focusing it on the current task. Failure- or threat-related action orientation refers to the ability to reduce the negative affect elicited by an aversive experience (Kuhl, 1994a).

Decision-related state orientation refers to the tendency to hesitate in decision making and to postpone actions under demanding conditions. Decision-related action orientation describes individuals who make decisions and initiate difficult actions relatively quickly under demanding or difficult conditions (Kuhl, 1994a; see also Chapter of Ruigendijk, Jostmann, & Koole, this volume). The performance-related state orientation refers to individuals who have the need of switch to a different activity even after doing well on the current activity. The performance-related action orientation describes the tendency to remain in an activity until its completion without the need of changing to another activity (Kuhl, 1994a). Whereas the hesitation observed in state-oriented individuals in the decision dimension suggests an under-functioning of the action initiation system, the volatility observed in state-oriented in the performance dimension can be explained with an over-functioning of the action initiation system (Kuhl, 1994a, pp. 52-53).

The personality dimension of state versus action orientation has generated a big amount of research. In the early 1990s, an entire volume with 31 chapters was devoted to this personality dimension (Kuhl & Beckmann, 1994), and the literature on action-state orientation has grown considerably since then (for more recent reviews, see Chatterjee, Baumann, & Osborne, 2013; Diefendorff, Hall, Lord, & Strean, 2000; Koole, Jostmann, & Baumann, 2012; Kuhl & Baumann, 2002). Most papers focus specifically on one type of state versus action orientation. Studies dealing with the threat-related dimension have investigated "learned-helplessness" (Kuhl, 1981), the degenerated intention hypothesis of depression (Kuhl & Helle, 1986), or the "self-infiltration" effect (Baumann & Kuhl, 2003; Baumann, Kuhl & Kazén, 2005; Kazén, Baumann, & Kuhl, 2003; Kuhl & Kazén, 1994; Quirin, Koole, Baumann, Kazén, & Kuhl, 2009). Studies dealing with the decision-related dimension, have investigated prospective memory (Goschke & Kuhl, 1993; Kaschel, Kazén, & Kuhl, 2016; Kazén, Kaschel, & Kuhl, 2008) or deficits in volitional action (Heckhausen & Strang, 1988; Jostmann & Koole, 2007; Gröpel & Kazén, 2014; Koole & Jostmann, 1994; Koole, et al., 2012). The third dimension, volatility, has been less well investigated, but there is evidence of its relevance to explain sport performance (Beckmann & Kazén, 1994), flow experiences (Baumann, Lürig, & Engeser, 2016; Keller & Bless, 2008), and global versus local processing styles (Marguc, Förster, & van Kleef, 2011).

Personality Systems Interactions (PSI) Theory

Kuhl's original theory of action control covered personality differences in motivation and volition. While motivation and volition are important aspects of personality, it is clear that personality functioning entails many other aspects, such as drives, habits, affect, coping and so on. From the 1990s and onwards, Kuhl worked to extend the action control perspective to include all aspects of personality functioning. This resulted in Personality Systems Interactions (PSI) theory (Kuhl, 2000, 2001, 2010).

PSI theory is a broad theoretical approach that seeks to explain all aspects of personality functioning in action-theoretical principles. Traditional personality theories focus only on one aspect of personality, such as drives, goals, or the self. PSI theory therefore represents a “macro-theory”, a general framework that integrates a number of well-known theories of personality that deal with more circumscribed aspects of personality. More specifically, PSI theory posits a hierarchical organization of personality functioning (Kuhl & Koole, 2008; Kuhl & Quirin, 2011). This organization is based on the degrees of freedom that each level affords: At the lowest level, people’s actions are determined by rigid stimulus-response links which afford little freedom, whereas at the highest level, people’s actions are determined by more abstract goals and values, which afford many alternative courses of action, and hence much more freedom in action control. This hierarchy is based on the notion that “the experiential and action-relevant processes of personality exhibit a hierarchical structure, which in the phylogenesis … have evolved from simpler to increasingly more complex functional levels” (Kuhl, 2010, p. 435).

PSI theory distinguishes seven levels of personality functioning that each can explain behavior in a characteristic way and each have been the focus of a major school of personality psychology. At the first personality level (*habits*), basic processes of intuitive behavioral control and object recognition are distinguished. This level has been traditionally the focus of behaviorists like Burrhus F. Skinner and (initially) Albert Bandura. At the second level (*temperament*), motor activation and sensory arousal are differentiated which have been studied by personality psychologists like Hans Eysenck (although he did consider temperament as a general source of energy). At the third level (*affect*), the influence of positive and negative affect on behavior and experience is described which has been the focus of theorists like Jeffrey Gray. The fourth level (*coping with stress*) is considered as an interface or “switch” between low- versus high-inferential processes. Under stress, it will be decided which levels of personality take control of behavior and experience: “Regression”

(habits, temperament, or affect) or “Progression” (motives, cognitive functions, or self-management). These coping processes have been the focus not only of Sigmund Freud but also of theorists such as Richard Lazarus.

At the fifth level (*motives*), social motives are represented (affiliation, achievement, power, and more recently, autonomy). The motive level has been the traditional focus of theorists such as David McCllelland, Heinz Heckhausen, and John Atkinson. At the sixth level (*cognitive functions*), analytical versus holistic cognitive processes are described. This level has been the focus of cognitive personality psychologists such as George Kelly. At the seventh level (*self-management*), self-control (or “discipline”; i.e., more “dictatorial”) and self-regulation (more “democratic”) are differentiated as alternative forms of self-management. This seventh level has been the focus of humanistic oriented theorists such as Carl Rogers or Edward Deci and Richard Ryan, and self-theorists such as Roy Baumeister.

At all personality levels, the focus of PSI theory is not on mental contents (e.g., expectations, self-ascriptions, intentions, goals, values) but rather on a functional perspective, by which behavior and experience is described according to the relative activation of mental systems (see Kuhl, 2001, Chapter 4). This approach is akin to that of the neurosciences, in which systems (neurobiological structures) are related to mental processes according to the functioning and relationships among them and not only through subjective experience. The functional approach has the advantage that the postulated relationships among systems can be applied to many different contents (e.g., work, religion, school work, sport, etc.). Details of the neurobiological basis of PSI theory are found in Kuhl (2001).

The personality hierarchy of PSI theory is an important contribution on its own, and can be used as a taxonomy of personality processes. But the theory goes well beyond the taxonomic level, by addressing the dynamic processes that operate between the different levels. Indeed, dynamic interactions between personality systems are the main focus of PSI theory. In the next section, we outline the principles that govern these interactions.

Dynamics of Personality Systems Interactions

According to PSI theory, the highest level, complex personality systems (the self, goals) does not directly make contact with the external world. Higher-level personality systems must therefore interact with lower level, elementary systems, in order to guide the individual's actions. Dynamic interactions between personality systems are thus essential for adaptive personality functioning. More specifically, PSI theory assumes that the interactions between two "low-inferential" (elementary) and two "high-inferential" (complex) personality systems are especially important. As a first step towards understanding these personality systems interactions, we first discuss each of the four personality systems in more detail. The functional profile of each system can be seen in Table 1. Their main features and interactions are shown in Figure 1.

Personality Systems

Intention Memory. The intention memory system is involved in the formation and maintenance of conscious intentions whenever an action cannot be carried out immediately through automatic behavioral programs (Goschke & Kuhl, 1993) and conveys a sequential-analytic form of action-readiness thought. The intention will remain in intention memory until a favorable situation for its enactment appears. The enactment of the intention in this situation is considered to be facilitated by positive affect (Kuhl & Kazén, 1999). Individuals with a fixation in intention memory have a tendency to think a lot about their intentions and ideals but do little for their attainment. Functional properties of this system are listed in Table 1, upper left panel.

Intuitive Behavior Control. The intuitive behavior control system is responsible for the execution of behavioral routines and the enactment of intentions. In agreement with intention memory, the intuitive behavior control system can translate an intention into action. The functional properties of intuitive behavior control are listed in Table 1, lower right panel. Intuitive behavior control is a parallel-processing system that integrates momentary

perceptual parameters that are sensitive to orientation, movement, and contextual information related to actions. It automatically processes the enactment-relevant information without the need of conscious action control. Intuitive behavior control supports a basic form of intuition. It can be activated through imitation, indirect cues (primes), and parameter specification (Bargh, Chen, & Burrows, 1996; Gollwitzer, 1999). If the motor schemas needed for action execution are well-learned, then the influence of intuitive behavior control is greater than that of intention memory on action control. That is, intuitive behavior control can act independently and even override activation of intention memory.

Object Recognition. The object recognition system focuses on single objects and isolates them from the entire context. It is specialized in detecting perceptual or conceptual information that is not congruent with our expectations or needs. This feature is particularly important to avoid sources of danger. It supports the sequential-analytic processing of perceptual objects. Object recognition detects in relationship with negative affect whether the results of an action or other perceptual experiences are incongruent with our expectations or needs. Through a comparison between what is expected and what is perceived, it tends to focus on the past. If object recognition becomes dominant (e.g., by reduced control from other systems) it can lead to rumination about a given state, resulting in a fixation on objects or contents related to this state, for example, about a failure, an accident, or a painful experience. Individuals who over-activate the object recognition system can become “discrepancy experts”.

Extension Memory. The extension memory system integrates congruent and incongruent experiences in existent parallel networks of life experience and provides information about own needs, motives, values, and emotions related to those life experiences and to self-aspects. Extension memory is a high-level parallel processing system that permits creative task enactment and supports goal pursuit whenever the global context (overview) is taken into account, such as relevant autobiographical experience, action alternatives, one’s

own and external needs, up to the meaningfulness of a task. It mostly processes without explicit conscious awareness, although contents processed by it can be made conscious by intention memory. Extension memory can be considered an intelligent form of intuition. Not least, other than intention memory, information processed by extension memory is linked to emotions. In addition, extension memory is capable of integrating positive and negative experiences to form new schemas and to provide meaning for an individual.

Modulation Assumptions

Individual differences in personality are explained in PSI theory according to (a) differences in activation threshold (dominance) of the four systems and (b) through the strength of the individual connections among the systems, that is, the effectiveness by which they interact (see Figure 1 and Table 1). Detailed information about individual differences in personality, that is, about personality styles and disorders from the perspective of PSI theory, can be found in Kuhl (2001, Chapter 16) and in Kuhl and Kazén (2009).

The interactions between the personality systems are explained by a separate set of principles, the so-called modulation assumptions of PSI theory. Common to both modulation assumptions is the notion that affective changes are vital for achieving personality systems interactions. The interaction of antagonistic systems succeeds only through affective change or “emotional dialectic” (Kuhl, 2000, 2001). The affect builds in some measure the “engine” which drives the communication between systems and in this way determines both behavior and experience.

The first modulation assumption is about positive affect, which is assumed by PSI theory to modulate the interaction between intention memory and intuitive behavior control. In general, positive affect (e.g., feelings of happiness) results when a desired goal is attained or when one makes steps to approach the goal (Carver & Scheier, 2002). If positive affect is present, action will be controlled by automatic routines of intuitive behavior control. It can facilitate intuitive, spontaneous, and creative behavior (Bledow, Rosing, & Frese, 2013; Bolte,

Goschke, & Kuhl, 2003). On the other hand, positive affect is inhibited if approaching the goal becomes more difficult than expected (frustration), with the consequence that intention memory is activated to store and maintain the goal. The inhibition of positive affect that results from activation of intention memory with a difficult intention inhibits intuitive behavior control. This inhibition of intuitive behavior control has the purpose of disconnecting thought from action, which allows the individual to generate a feasible action plan and to resolve the anticipated difficulties through analytical thinking. This could take the form of thinking carefully about each step to reach the goal. When a satisfactory solution to the problem is found, the resulting positive affect will help to reconnect the inhibited connection between intention memory and intuitive behavior control and initiate (volitional) action. The first modulation assumption has been directly confirmed by a number of well-controlled experiments (Kazén & Kuhl, 2005; Kuhl & Kazén, 1999; see also Baumann & Scheffer, 2010).

The second modulation assumption relates to negative affect and the interaction between extension memory and object recognition. It aims to explain self-growth (see Figure 1). The object recognition system is activated by negative affect (fear, anxiety, helplessness, or sadness). The activation of object recognition blocks extension memory, with the consequence that the individual loses the overview of the situation and focusses on isolated percepts or memories. On the other hand, activation of the self-representations of extension memory in threatening situations leads to self-reassurance (*Selbstberuhigung*) and reduction of negative affect. This reduction is more effective if the connection between extension memory and mechanisms related to affect regulation is strong (between right-frontal cortical structures and hippocampus; see Mayberg et al., 1999). In coping with stress, activation of extension memory generates a “high-inferential” overview of our life experiences in which alternative solutions are searched for, replacing a narrow view of a problem (“tunnel vision”). The activation of extension memory is also a precondition for the process by which

unexpected, undesired, or painful experiences perceived by the object recognition system are integrated into extension memory. In this way, extension memory become more and more differentiated and complex ("personality growth"). This process can be the basis by which the "narrative method" therapy works, which is applied among others to patients with post-traumatic stress disorder (see e.g., Pennebaker & Chung, 2011).

Two Illustrations of the Integrative Potential of PSI theory

PSI theory has been developed as an integrative framework for understanding motivation, volition, and personality processes. It would go far beyond the present context to discuss all the integrative applications of the theory. We therefore limit ourselves to two topics to illustrate PSI theory's integrative potential: The Rubicon model and individual differences in action versus state orientation. We chose these topics because of their historical relevance to the development of PSI theory.

The Rubicon Model and PSI Theory

The Rubicon model from Heckhausen and Gollwitzer (1987), which built on Kuhl's early work (1983; Hechkausen & Kuhl, 1985), has nowadays become one of the mostly recognized models of motivation and volition. To recapitulate, the Rubicon model postulates four phases that individuals run through when they want to perform a certain action: Pre-decisional, pre-actional, actional, and post-actional, which are proposed to explain, respectively: How do people select their goals? How do they plan the execution of those goals? How do they enact those goals? And how do they evaluate their efforts to accomplish a specific goal?

Although the Rubicon model is anchored in the phenomenological experience of the transition from weighing to willing, it can still be described in terms of personality systems interactions: PSI theory assumes that during each of the four phases one of the cognitive personality systems is predominantly active. Specifically, the pre-decisional phase (weighing of preferences), should be carried out within the extension memory system. This phase is

motivational and extension memory is closely tied to motivational and emotional systems, mostly in an implicit way, and it is also involved in affect regulation, which may be needed to increase the attractiveness of a boring intention or to reduce the displeasure of a difficult intention. The outcome of this phase should be a commitment to carry out the intention, to "cross the Rubicon". The second, pre-actional phase should engage the intention memory system, in which planning takes place, and specific steps to carry out the intention are systematically considered. The third, actional phase should engage the intuitive behavior control system, once the specific parameters governing the action are set. If the intention is difficult it may require positive affect to overcome the inhibition that intention memory exerts on intuitive behavior control (cf. Kuhl & Kazén, 1999). Finally, the fourth, post-actional phase should engage mainly the object recognition system, in which a critical evaluation of the action takes place (What does not fit?), but also in close exchange with extension memory (What does fit?).

By adopting such an approach, interactions between these systems during goal pursuit can be described. In fact, rather than postulating a clear sequence of phases, PSI theory suggests the possibility that systems that are predominantly active in one phase can also support the processes in another (e.g., subsequent) phase. For example, whereas extension memory (and the self) provides a circumspect overview of self-aspects and preferences during the pre-decisional phase for making a good choice, it may become activated during action initiation in order to provide motivation (e.g., feelings of self-efficacy and meaningfulness) in order to stay on track. Also, if the chosen goal is not congruent to the individual's self-aspects anymore or the chosen goal becomes unlikely to be reached, reactivation of extension memory including updated preferences and chances helps the individual to disengage from the goal (e.g., Brandstätter & Herrmann, 2016; see also Brandstätter & Herrmann, chap. Xx, this volume).

In addition, PSI theory attaches much importance to the interaction between object recognition (predominant in the evaluation phase) and extension memory (predominant in the weighing phase) as a basis of "learning from mistakes", and thus combines the issue of action control (or self-regulation, in a narrow sense) with the developmental issue of self-growth within a single model. For example, evaluations and failure experiences as represented locally by object recognition system at first can become integrated into the self and thus add to a more profound basis for decision-making in future, similar situations.

PSI Theory and Emotion Regulation

During the last decades, the concept of emotion regulation has attracted much scientific attention and has been the focus of an enormous number of empirical studies (Gross, 2015). Emotion regulation refers to changing the course of an affective response to adjust oneself to external or personal demands, standards, or goals (e.g., Gross, Sheppes, & Urry, 2011). Accordingly, emotion regulation can be conceived of as a process that plays a central role in self-regulation. In fact, this notion is reflected in PSI theory's modulation assumptions, according to which emotion regulation facilitates the information transfer between personality systems for the purpose of action control. Specifically, it is considered that individual differences in action versus state orientation are related to the ability to regulate emotions (Kuhl & Beckmann, 1994; see also Baumann, Kaschel, & Kuhl, 2007; Quirin, Düsing, Kuhl, 2011): Whereas threat-/failure-related action orientation corresponds with the regulation of negative affect, decision-related action orientation corresponds with the regulation of positive affect. According to PSI theory, regulation of negative affect versus positive affect have different implications for personality functioning.

Regulation of negative affect. The regulation of negative affect modulates the interaction between extension memory and object recognition. Whereas state-oriented individuals have impairments in down-regulating negative affect under threatening conditions, action-oriented

individuals are able to down-regulate negative affect through activation of extension memory, which allows them to have an overview of the situation and to avoid “tunnel vision” (see Figure 2.1). A good access to one’s own preferences, values, emotions, and experiences within extension memory is a precondition for a meaningful life, in which self-congruent goals are pursued and lead to satisfaction when they are reached (Ryan & Deci, 2000).

On the other hand, if negative affect is present, object recognition will be activated and access to extension memory will be blocked initially. Impaired access to extension memory can be described as “latent alienation”: individual do not know what they want (see Figure 2.1). The latter occurs especially for threat-related state orientation under aversive conditions. Individuals may experience latent alienation as negatively toned ruminations that they cannot stop voluntarily. Impaired access to the self also occurs under external pressure, implicit or explicit. Under aversive conditions, state-oriented individuals become alienated from their own personal preferences, rely on implicit or explicit suggestions of others in goal pursuit, and show a tendency towards “self-infiltration”, that is, they misattribute externally suggested goals or norms as self-generated (Kuhl & Kazén, 1994; see also Baumann & Kuhl, 2003; Baumann, Kuhl & Kazén, 2005; Quirin et al., 2009). Self-infiltration is not observed in action-oriented individuals under aversive conditions, presumably because they can access their own preferences through a process of autonoetic access (Kazén et al., 2003). Individuals may also be able to finally regain access to blocked extension memory under negative conditions by using self-regulatory strategies postulated by action control theory such as selective attention, encoding control, or environment control (see above; see also Kuhl, 1984; Gross, 2015).

Regulation of positive affect. Decision-related action-state orientation is related to positive affect regulation, which modulates the interaction between intention memory and intuitive behavior control. According to PSI theory, state-oriented individuals have difficulties to up-regulate positive affect under boring or difficult conditions. These individuals hesitate at the

cognitive level and hesitate at the behavioral level and procrastinate, that is, they do not carry out a prospective task, even though they have the need, the opportunity, and the means to do it (e.g., Beswick & Mann, 1994, Blunt & Pychyl, 1998). Action-oriented individuals, by contrast, are able to self-generate positive affect through activation of extension memory (recall that the self-management process that involves emotions in the service of action control, has been called “self-regulation” in contrast to “self-control/discipline”). As a network that provides a broad overview of many contexts and self-aspects, extension memory can then provide meaning, action alternatives, and positive affect that allows to remove the volitional inhibition put on intuitive behavior control by intention memory (see Figure 2.1).

According to the first modulation assumption of PSI theory, efficient volitional action occurs through emotional dialectic on the positive affect axis that fosters a continuous exchange between intention memory and intuitive behavior control. Efficient volitional action control can be trained through techniques that foster emotional changes such as the mental contrasting method investigated by Oettingen, Pak, and Schnetter (2001). Many studies have demonstrated that state-oriented individuals, despite maintaining representation of their intentions more active than action-oriented individuals, still have difficulties enacting their intentions (Goschke & Kuhl, 1993; Goschke & Bolte, chap. 7, this volume; Kaschel et al., 2016; Penningroth, 2011; Ruigendijk, Jostmann, & Koole, chap. 9, this volume). Action-oriented individuals, by contrast, are able to self-motivate themselves and are more efficient in both decision-making and action initiation (Kazén et al., 2008), even in old age (Kaschel et al., 2016; Kaschel & Kazén, chap. 8, this volume). Moreover, compared to state-oriented, action-oriented individuals are less affected by the so-called “ego-depletion” effect, that is, they show less performance decrements on a second task after a resource-demanding first task (Gröpel, Baumeister, & Beckmann, 2014) presumably because they are better capable of drawing upon self-regulation (as compared to self-control), that is, the motivating functions of extension memory (Kazén, Kuhl, & Leicht, 2015).

Conclusions and Outlook

In the present chapter, we have reviewed the theoretical roots of PSI theory in the areas of motivation and volition, including individual differences in volitional action control. As we have seen, motivation and volition are an integral part of PSI theory but this integrative theory goes well beyond these topics by describing not only seven levels of personality but also its structural components (extension memory, intention memory, object recognition, and, intuitive behavior control) and the dynamics of their interaction through positive and negative affect.

Despite the apparent complexity of PSI theory, as reviewed here, the integrative power of this theory more than compensates the initial effort invested in learning some new concepts and thinking in functional (system activation) ways. It is certainly more difficult to learn and bring coherently together every isolated theory proposed to account for a great number of psychological phenomena in personality, motivation, social, cognitive, and clinical psychology. Actually, the most difficult task is not to learn the basic concepts of PSI theory but to apply them to the diversity of psychological phenomena that PSI theory can explain. Some of them have been mentioned here such as learned helplessness, ego-depletion, depression, trauma therapy, self-infiltration, automaticity, goal enactment, volitional facilitation, or prospective memory. The list of phenomena to which this theory can be applied can be easily continued.

Confirming a phrase attributed to Kurt Lewin that “there is nothing more applied than a good theory”, we want to finish this chapter by mentioning that PSI theory has already a wide range of applications in the most diverse areas such as clinical, organizational, and educational psychology since more than 15 years. PSI theory has stimulated the development of a diagnostic system, “Evolvement-Oriented System Diagnosis” (EOS), that measures a series of specific psychological processes, including motives (explicit and implicit), affect (explicit and implicit), personality and cognitive styles, self-government (self-regulation &

self-control), volition (implicitly), stress, and symptoms (see www.impart.de). Kuhl, Kazén, and Koole (2006) give an overview of some applications of PSI theory. The authors also describe three basic steps in the application of PSI theory: (a) *Diagnosis* through the EOS system, to find out which of the many psychological processes assessed show strengths or weaknesses. (b) *Intervention:* After identifying specific problems, PSI theory suggests ways to improve psychological functioning. According to PSI theory, primary responses to events (emotionality, cognitive styles) can be counteracted by self-regulation abilities (self-motivation and self-reassurance) and, thus, reduce the level of stress and symptoms. (c) *Evaluation:* Practitioners can examine expected changes in the individual's functions that turn from deficits before the intervention into behavioral changes and improvements in psychological and physical well-being after the intervention.

In our view, the best way to honor the contributions of Julius Kuhl to scientific psychology is to further explore, develop, and apply PSI theory in the many diverse academic and professional areas in which we work.

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Table 2.1.

Functional Profile of the Four Mental Systems (according to Kuhl, 2001)

Level	**Analytical** (left hemispheric)	**Holistic** (right hemispheric)
Complex	Intention Memory (IM)	Extension Memory (EM)
	- sequential-analytic	- parallel-holistic
	- slow impementation	- fast implementation
	- fast learning	- slow learning
	- explicit knowledge	- implicit configurational knowledge
	- either-or character	- integration of opposites
	- isolation of emotions	- perception & regulation of emotions
	- conscious	- not conscious
	- goal-focused attention	- congruence-related attention
Elementary	Object Recognition (OR)	Intuitive Behavior Control (IBC)
	- descontextualized	- contextualized
	- past centered	- present- and future-oriented
	- categorical	- prototypical
	- conscious	- not conscious
	- discrepancy-focused attention	- intuitive programs

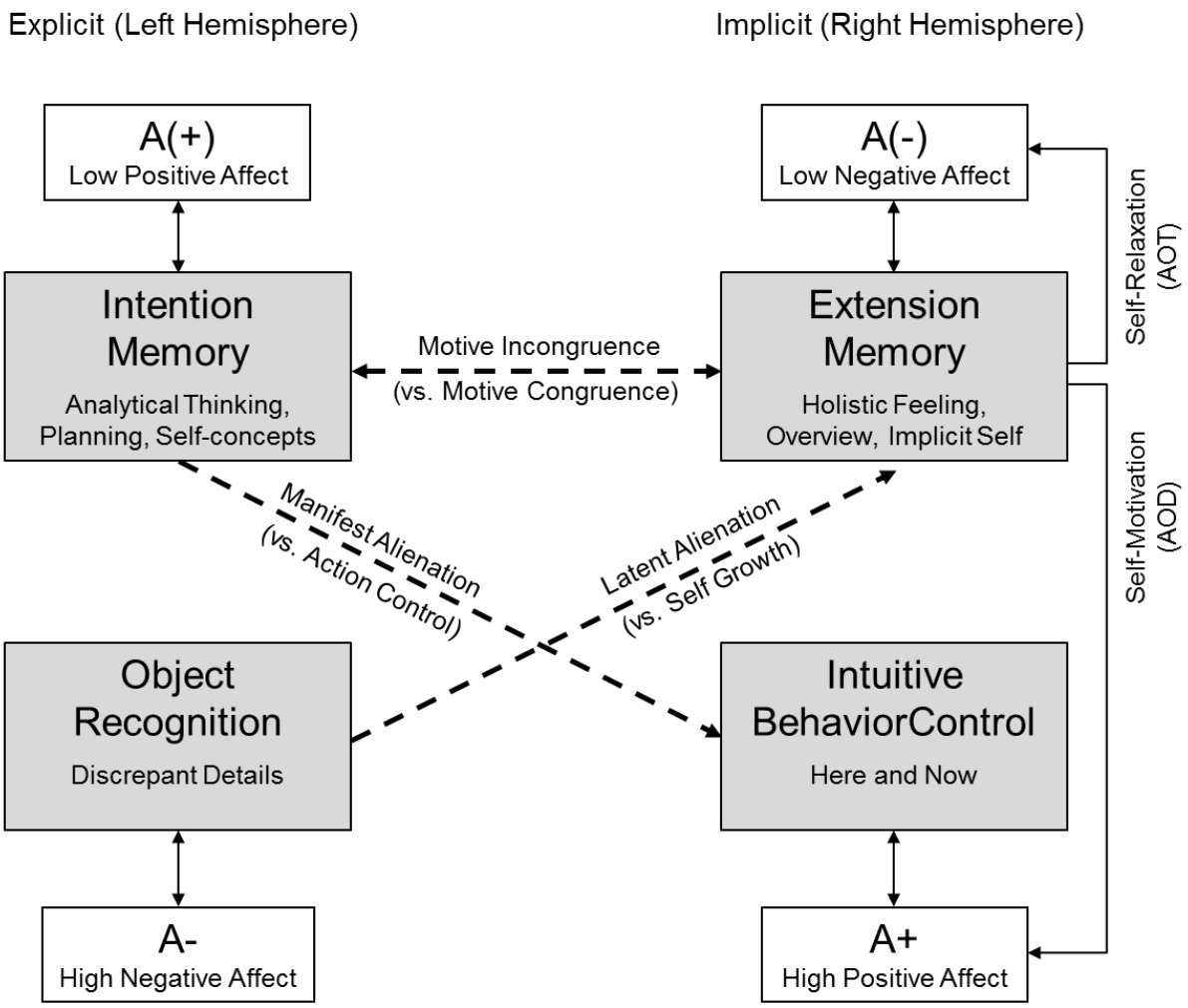

Figure 2.1. Cognitive systems of Personality Systems Interaction (PSI) theory and their modulation by affect (adapted from Kuhl, 2011). Note: Dashed arrows indicate antagonisms between cooperating systems. Affective changes from low to high positive affect (facilitated by demand-related action orientation, AOD) reduce manifest alienation and foster action control. Affective changes from high to low negative affect (facilitated by threat-related action orientation, AOT) reduce latent alienation and foster self-growth. Crosstalk between intention and extension memory fosters motive congruence. Self-motivation (AOD) helps to deactivate an overly strong intention memory under demanding conditions. Self-relaxation (AOT) helps to (re)activate extension memory under threatening conditions.